

# Errors

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*Failure Aesthetics*

# Glitch

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*"The glitch is a wonderful experience of an interruption that shifts an object away from its ordinary form and discourse."*

– Rosa Menkman

*A rupture in the seamless flow of digital operation*

*A revelation of the underlying processes and protocols*

*A critique of the belief in technological transparency and reliability*

*An aesthetic that foregrounds material failure rather than expressive intent*

*A method for engaging systems at their limits*

*A reminder that control in digital systems is always partial*

# Glitch

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post-digital: what comes after the novelty of digital tools, the belief in transparent, perfect sound, the ideology of “clean” digital audio.

- Failure shifts from technical problem to aesthetic and analytical condition
- Artifacts expose the assumptions embedded in digital tools and protocols
- Glitch reveals systems as operational, contingent, and historically situated
- Post-digital practice listens to failure rather than celebrating smoothness



# Data Bending

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**Data bending** is a glitch approach in which digital audio files are deliberately treated as **raw data** and **misinterpreted** by the wrong software or process. Instead of preserving the intended encoding, the file is opened, edited, or processed in a non-audio context (for example, a text editor or image editor), then played back as sound.

The resulting audio exposes artifacts caused by corrupted data structures: abrupt discontinuities, bursts of noise, timing errors, and unpredictable changes.

Data bending foregrounds the **fragility of digital representations**, treating errors and system misuse not as faults but as compositional material.

# Glitch Practice: Approaches

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## ***Accidental***

The attention and patience to notice and capture spontaneous system errors as they occur.

## ***Provocative***

Force glitches to emerge by pushing systems to their limits through noise, feedback, and incompatibility.

## ***Surgical***

Deliberately corrupt or manipulate data at the byte, code, or physical level to create controlled distortions.

## ***Generative***

Engineer unstable systems and code designed to produce autonomous, "misbehaving," or random outputs.

# Return to Modernism

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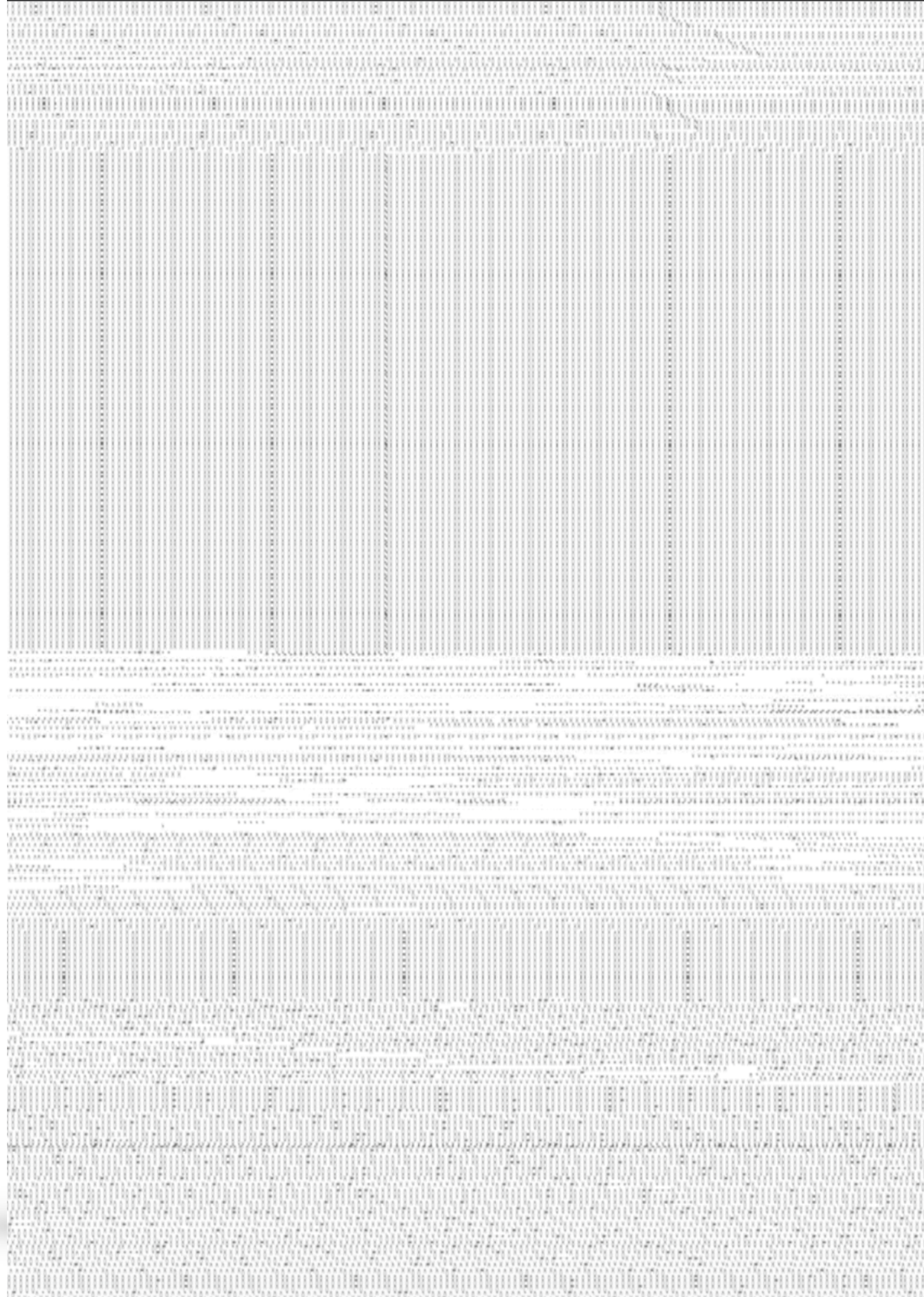
**Post-Digital:** A rejection of digital "purity" in favor of glitches, errors, and artifacts. It values the "noise" of the system as the primary content.

**Neo-Modernist Shift:** A reaction against Post-Modern "irony" and media saturation. Return to the purity and singularity of Modernism.

*For example:* Bare mathematical structures, minimalism, focus on the material substrate (skipping CD or overloaded processor).

*To discover the ontology (essence) of the medium, stripping away cultural references to reveal the "material ground."*





# Kim Cascone: The Aesthetics of Failure

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"The post-digital aesthetic was born from the detritus of the digital revolution."

– Cascone

**Post-digital:** Beyond digital novelty; now we examine its residue

**Background / foreground:** The "noise" of digital becomes the content

**Microsound:** Attention to the granular, the barely perceptible

**Failure as a portal:** Errors open doors to new aesthetics

"Artists were not rejecting digital technology; they were using it against itself."

# Rosa Menkman: The Glitch Moment(um)

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*The "glitch moment" is that instant of rupture when something breaks and reveals itself. The "(um)" adds momentum: the glitch doesn't just happen, it moves, it has consequences, it generates.*

A glitch only exists while it **disrupts** expectation. Once *stabilized*, it stops functioning as **glitch**.

1. It defines glitch as a moment, not a material
2. It frames glitch as revelatory, not decorative
3. It introduces time as the key axis of critique

# The Glitch Moment(um)

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The "*glitch moment*" has a particular structure:

EXPECTATION → RUPTURE → REVELATION → RESPONSE

**Expectation:** A system operates normally; we know what to expect

**Rupture:** Something goes wrong; expectation is violated

**Revelation:** Error shows something, about the system, about ourselves

**Response:** We decide what to do with this knowledge



# Rosa Menkman: Glitches

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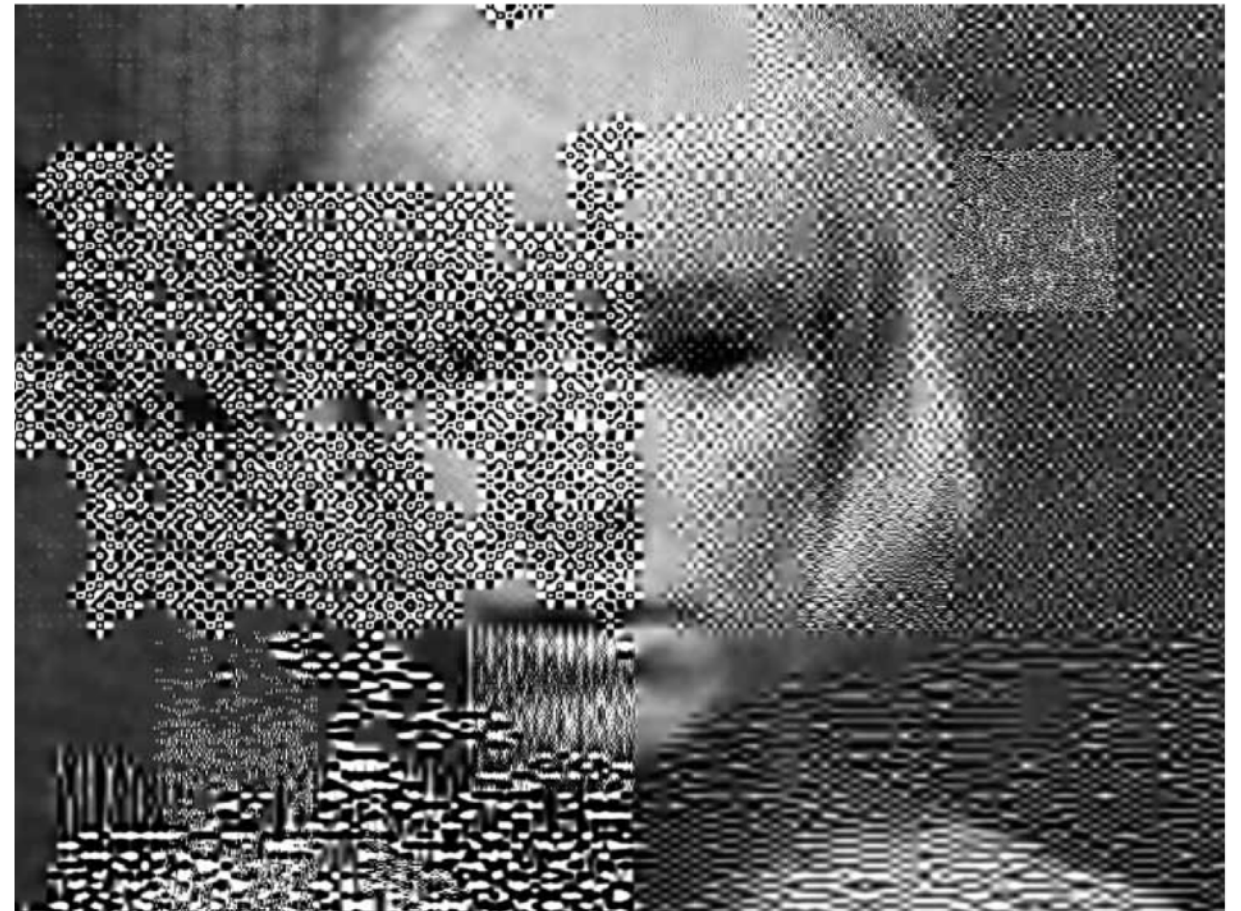
**A "cool" glitch** is the authentic, unstable moment of rupture. It is "cool" because it is low-definition in terms of "ready-made" meaning—it requires the viewer to work to understand it.

**A "hot" glitch** is what happens when the radical break becomes a "look." It is "hot" because it is high-definition; it gives you all the information you need and doesn't ask for much mental involvement.





A DATABENT BASELINE JOINT PHOTOGRAPHIC EXPERTS GROUP IMAGE (.JPEG).  
THIS BASELINE JPEG IMAGE WAS DATABENT BY THE INTRODUCTION OF A RANDOM ERROR TO THE INFORMATION DATA. THE IMAGE  
GENERATED IS AN IRREVERSIBLE DATABEND. THE JPEG COMPRESSION IS LOSSY.



**A DATABENT JOINT PHOTOGRAPHIC EXPERTS GROUP COMMITTEE 2000 IMAGE (.JPF).** | THIS JPEG 2000 IMAGE WAS DATABENT BY  
THE INTRODUCTION OF A RANDOM ERROR TO THE INFORMATION DATA. THE IMAGE GENERATED IS AN IRREVERSIBLE DATABEND. THE  
JPEG 2000 COMPRESSION IS LOSSY.

# Mitchell Whitelaw: The Grain of the Digital

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*"Inframedia: Glitches: clicks, pops and CD-skips... digital aliasing and the sharp clicks caused by discontinuities in a digital waveform."*

(Whitelaw 2004)

**Digital materiality:** Despite "immateriality," media has texture, grain, weight

**Compression artifacts:** The signature of compression are material traces

**Generative systems:** processes that produce unexpected outputs

"The primary concern here is not representation, but self-reflection on the system and processes of the media."

# Thor Magnusson: Processor Art

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*"Processor art concerns works that necessarily must run on the computer in order to exist."*

The **artwork** is:

- the executing process
- the rules
- the temporal unfolding

Any **autonomous process**:

- can diverge
- can destabilise
- can produce unintended behaviour

# Yasunao Tone: Wounded CD

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**The wounded:** Physically damaging the carrier transforms the content

**Stochastic processes:** Introducing randomness into deterministic systems

**Anti-perfectionism:** The CD was marketed as "perfect sound forever"

**Error as revelation:** sounds as traces of error-correction and buffering

**Material critique of the digital:** digital perfection relies on fragile ground

*"the prepared CD seldom repeated the same sound when I played it back again, and it was very hard to control"*

*"To fight with smart machines you have to be very primitive"*

# Richard Chartier: Removal

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*"My work is really a process of removal. Sometimes a piece will be based on one [looped] sound with things layered over it, and then eventually I will take the original linking element out. So it's this ghost element that's not really there. That absence continues to exert a pressure on the work, lurking as an implied presence in the tracks and compounding the obscure impression that something is missing."*

The **organizing principle** is **blurred** by removing important initial material.

Highlights **later choices** as the process unfolds.





# Waveset Distortion

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A digital sound-processing technique that operates on **wavesets**, short waveform segments defined by **zero-crossings**.

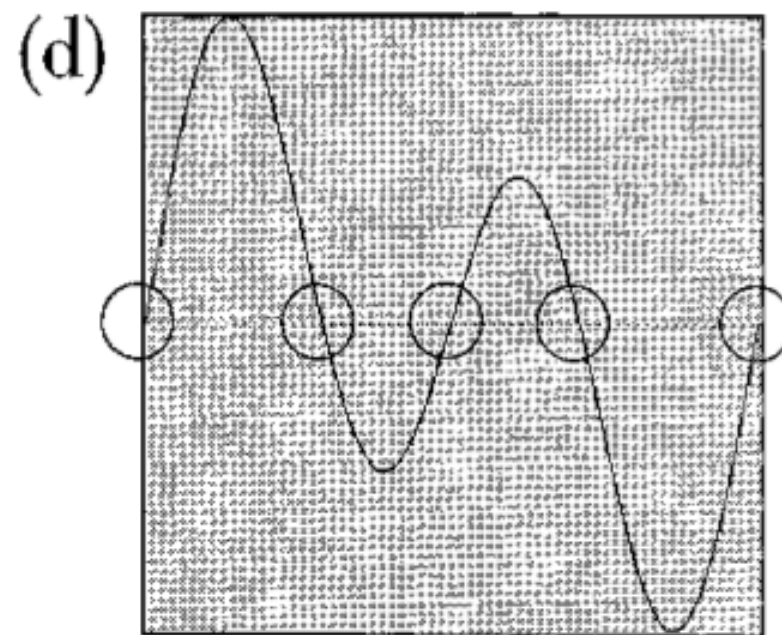
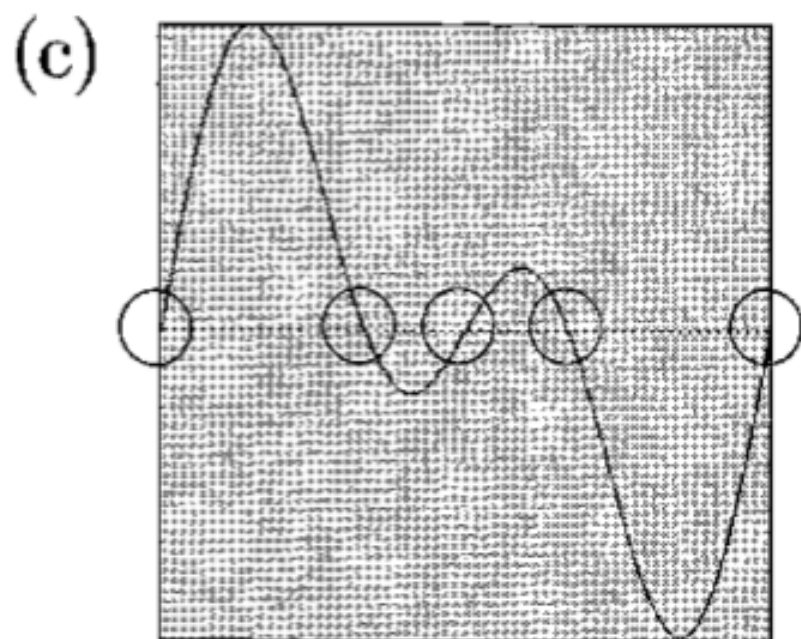
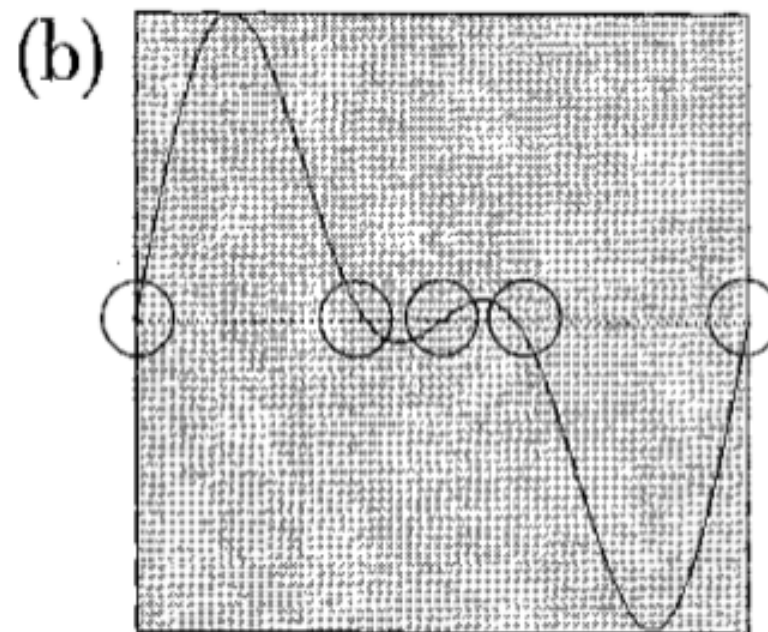
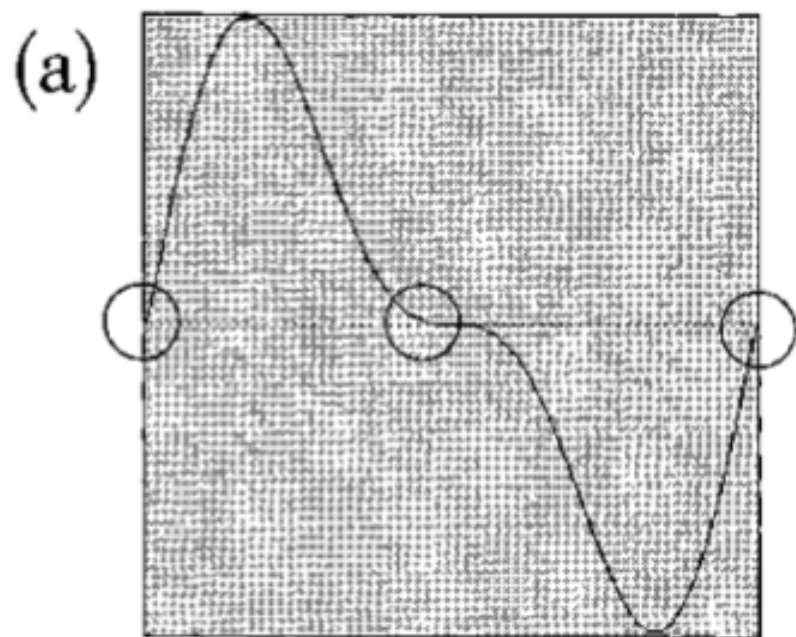
The signal is segmented and the **wavesets** are reordered, repeated, inverted, removed, or transformed, producing timbral changes that range from subtle spectral shifts to aggressive, **glitch-like distortion**.

Used extensively in **Trevor Wishart's** CDP tools and works such as *Tongues of Fire*, it exposes and reshapes the microscopic structure of sound.



# Wishart

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## Hecker - Stocha Acid Zlook (2001)



*Pandämonium 9 Playlist and Stocha Acid Vlook are hybrids that are equally fed from the Xenakian Dynamic Stochastic Synthesis or GENDYN <sup>2</sup> for short – a »Music out of Nothing« as Peter Hoffman recently termed it – and a structuring, processing and editing approach to sound units that was conceived by the composer Trevor Wishart, in units which he calls »Wavesets« <sup>3</sup>. The pieces clearly celebrate these techniques and are, to some extent, »improvements« on these tools which function as means of producing intensity.*

```

(
  NF(\iop, {|freq=78, mul=1.0, add=0.0|
    var noise = LFNoise1.ar(0.001).range(freq, freq + (freq * 0.1));
    var osc = SinOsc.ar([noise, noise * 1.04, noise * 1.02, noise * 1.08],0,0.2);
    var out = DFm1.ar(osc,freq*4,SinOsc.kr(0.01).range(0.92,1.05),1,0,0.005,0.7);
    HPF.ar(out, 40)
  }).play;
)

(
  NF(\dsc, {|freq = 1080|
    HPF.ar(
      BBandStop.ar(Saw.ar(LFNoise1.ar([19,12]).range(freq,freq*2), 0.2).excess(
        SinOsc.ar( [freq + 6, freq + 4, freq + 2, freq + 8])),
        LFNoise1.ar([12,14,10]).range(100,900),
        SinOsc.ar(20).range(9,11)
      ), 80)
    ).play;
)

var <>pindex, <>cindex;

initialize {
  if(pindex.isNil, { pindex = 1000 });
  if(cindex.isNil, { cindex = 2000 });
}

clearProcessSlots {
  pindex = 1000;
  (this.pindex - 1000).do{|i| this[this.pindex+i] = nil; }
}

clearOrInit {|clear=true|
  if(clear == true, { this.clearProcessSlots() }, { this.initialize() });
}

transform {|process, index|
  if(index.isNil && pindex.isNil, {
    this.initialize();
  });

  pindex = pindex + 1;
  this[pindex] = \filter -> process;
}

control {|process, index|
  var i = index;

  if(i.isNil, {
    this.initialize();
    cindex = cindex + 1;
    i = cindex;
  });

  this[i] = \pset -> process;
}

(
  NF(\depfm, {|freqMin=5, freqMax=20, mul=20, add=80, rate=0.5, modFreq=2100, index=0.3, amp=0.2|
    var trig, seq, freq;
    trig = Dust.kr(rate);
    seq = Diwhite(freqMin, freqMax, inf).midicps;
    freq = Demand.kr(trig, 0, seq);
    HPF.ar(PMOsc.ar(LFCub.kr([freq, freq/2, freq/3, freq/4], 0, mul, add),
      LFNoise1.ar(0.3).range(modFreq,modFreq*2), index) * amp, 50)
  }).play;
)

```

## Exercises

# Exercises

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## **Exercise 1: Pattern to Buffer**

Use the `~patternToBuffer` function to create a buffer from a `Pbrown` pattern.  
Play it back at different speeds using `MouseX` to control the rate.

## **Exercise 2: Fold Distortion**

Create a `SinOsc` and fold it using `.fold2` with a modulated threshold.  
Use `LFNoise2` to slowly change the fold amount between 0.1 and 0.9.

## **Exercise 3: Signal Comparison**

Use the `<` (less than) operator to compare a Saw wave with a `SinOsc`.  
Wrap the result in `LeakDC.ar` to remove the DC offset.  
Experiment with different frequencies for each oscillator.

## **Exercise 4: Sample Rate Reduction**

Use `Latch.ar` and `Impulse.ar` to reduce the sample rate of a signal.  
Control the sample rate with `MouseX` between 500 and 10000 Hz.

## **Exercise 5: Waveset Time Stretch**

Modify the waveset playback to repeat each waveset multiple times.  
Use `\repeats` to stretch the sound in time.

# Exercises

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## **Exercise 6: Random Waveset Selection**

Instead of playing wavesets sequentially, use Pwhite to randomly select which waveset to play.

## **Exercise 7: Direction Scrambling**

Use PlayBuf with a rate that randomly switches between positive and negative values using LFNoise0.

## **Exercise 8: Instruction Synthesis**

Create a buffer with random values and use the Instruction UGen to synthesize sound from it. Experiment with different buffer sizes.  
Note: Requires sc3-plugins to be installed.

## **Exercise 9: LPCError Texture**

Use LPCError.ar with a noise input and modulated prediction order. Filter the result with BPF to tame the harshness.

